

Trust-Aware AI-Assisted Service Placement in Edge Computing Systems

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Trust-Aware AI-Assisted Service Placement in Edge Computing Systems

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in

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by

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based on research carried out

under the supervision of

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March 05, 2026

Supervisor's Certificate

This is to certify that the work presented in the project report entitled *Trust-Aware AI-Assisted Service Placement in Edge Computing Systems* submitted by *Pediredla Surya Venkata Mourya*, Roll Number 122CS0563, is a record of original research carried out by him under my supervision and guidance in partial fulfillment of the requirements of the degree of *Bachelor of Technology in Computer Science and Engineering*. Neither this project report nor any part of it has been submitted earlier for any degree or diploma to any institute or university in India or abroad.

Dr. Bibhudatta Sahoo

Dedication

This report is dedicated to my family and mentors for their constant support, motivation, and guidance throughout my academic journey.

Their encouragement has been the driving force behind my learning and completion of this research work.

Pediredla Surya Venkata Mourya

Declaration of Originality

I, *Pediredla Surya Venkata Mourya*, Roll Number *122CS0563* hereby declare that this project report entitled *Trust-Aware AI-Assisted Service Placement in Edge Computing Systems* presents my original work carried out as a undergraduate student of NIT Rourkela and, to the best of my knowledge, contains no material previously published or written by another person, nor any material presented by me for the award of any degree or diploma of NIT Rourkela or any other institution. Any contribution made to this research by others, with whom I have worked at NIT Rourkela or elsewhere, is explicitly acknowledged in the dissertation. Works of other authors cited in this dissertation have been duly acknowledged under the sections “Reference” or “Bibliography”. I have also submitted my original research records to the scrutiny committee for evaluation of my dissertation.

I am fully aware that in case of any non-compliance detected in future, the Senate of NIT Rourkela may withdraw the degree awarded to me on the basis of the present dissertation.

March 05, 2026
NIT Rourkela

Pediredla Surya Venkata Mourya

Acknowledgment

The process of completing this research project has been a valuable learning experience that allowed me to explore concepts related to edge computing, service placement strategies, and system reliability in distributed environments. Through this work, I was able to understand how intelligent decision mechanisms can improve the performance and dependability of edge-based systems.

I would like to express my sincere gratitude to my supervisor, **Dr. Bibhudatta Sahoo**, for his continuous guidance, valuable suggestions, and encouragement throughout this project. His insights and support played an important role in shaping this work and improving my understanding of the research process.

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Finally, I would like to express my heartfelt gratitude to my family for their constant encouragement, patience, and support throughout my academic journey.

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Abstract

Edge computing has become an important paradigm for supporting latency-sensitive applications such as Internet of Things (IoT), smart transportation, and real-time monitoring systems. In such distributed environments, computational services are deployed across multiple edge nodes rather than centralized cloud servers. Selecting reliable nodes for executing tasks therefore becomes a critical challenge.

This work presents a trust-aware service placement approach for edge computing environments. The proposed system evaluates the reliability of edge nodes using performance indicators such as trust score, latency, and historical task success rate. A machine learning model is used to estimate the reliability of nodes and assist in selecting suitable nodes for service deployment.

Simulation experiments are conducted to compare three placement strategies: random placement, trust-based placement, and AI-assisted placement. The results show that incorporating machine learning into the placement process improves task success rate and reduces failures when compared with baseline methods.

Keywords: Edge Computing, Service Placement, Trust Management, Machine Learning, Distributed Systems.

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Chapter 1

Introduction

1.1 Background

Edge computing has become an important paradigm for supporting applications that require low latency and real-time data processing. Instead of relying only on centralized cloud servers, edge computing allows computational tasks to be executed closer to where the data is generated, such as IoT devices, gateways, and local edge servers.

This architecture improves system responsiveness and reduces network traffic. However, the distributed and heterogeneous nature of edge environments introduces several challenges. Edge nodes may differ in terms of computational capability, reliability, and network conditions. As a result, deciding where services should be deployed becomes a critical problem.

If services are deployed on unreliable nodes, the system may experience task failures, higher latency, and inefficient resource utilization. Therefore, intelligent service placement strategies are required to evaluate node reliability and select appropriate nodes for executing tasks.

1.2 Project Objectives

The overall objective of this project is to design a reliability-aware service placement framework for edge computing environments.

The project focuses on the following objectives:

- To design a framework capable of monitoring edge node behavior and maintaining reliability information through trust metrics.
- To develop an intelligent placement mechanism that selects suitable edge nodes for executing services based on system metrics such as trust score, latency, and historical performance.
- To incorporate machine learning techniques to predict node reliability and assist in service placement decisions.

- To evaluate different placement strategies through simulation experiments and performance analysis.

1.3 System Overview

The proposed framework consists of several key components that work together to support reliability-aware service placement in edge computing environments.

Edge nodes represent distributed computational resources capable of executing tasks. Each node maintains statistics such as successful task executions, failures, and performance metrics.

A trust management mechanism tracks node behavior and updates trust scores based on task outcomes. Nodes that successfully execute tasks maintain higher trust values, while nodes with repeated failures experience trust reduction.

To maintain transparency and provide an auditable history of system events, a lightweight blockchain-based ledger is used to record important system activities such as task execution logs and trust updates.

A placement controller is responsible for selecting suitable nodes for executing tasks. The controller evaluates node trust scores and performance metrics in order to determine the most reliable node available.

In addition to trust-based evaluation, a machine learning model is introduced to analyze node performance data and predict the likelihood of successful task execution.

1.4 Experimental Approach

To evaluate the effectiveness of the proposed placement framework, simulation experiments were conducted using a prototype implementation.

A dataset representing edge node behavior was generated using system metrics such as trust score, success rate, latency, and resource usage. This dataset was used to train a machine learning model capable of predicting node reliability.

Three service placement strategies were evaluated:

1. Random placement, where tasks are assigned to nodes without considering reliability.
2. Trust-based placement, where nodes with higher trust scores are preferred.
3. AI-assisted placement, where machine learning predictions are used to select nodes expected to execute tasks successfully.

Experimental results indicate that AI-assisted placement improves task success rates compared to traditional placement approaches.

1.5 Future Work

While the current implementation demonstrates the feasibility of the proposed approach, further evaluation is required to analyze system behavior in large-scale edge environments.

Future work will involve integrating the proposed placement mechanism with the iFogSim simulation framework in order to evaluate performance under realistic workloads and large-scale network configurations.

1.6 Report Organization

The remainder of this report is organized as follows.

Chapter 2 reviews related research in edge computing, trust management, and service placement strategies.

Chapter 3 presents the architecture and methodology of the proposed framework.

Chapter 4 describes the implementation details and experimental results obtained from simulation studies.

Chapter 5 concludes the report and outlines future research directions.

Chapter 2

Literature Review

2.1 Overview

Edge computing has emerged as an effective paradigm for supporting applications that require low latency and real-time data processing. Instead of sending all data to centralized cloud infrastructures, computational tasks are executed closer to the data source at edge nodes. This approach significantly reduces network latency and improves system responsiveness for applications such as Internet of Things (IoT), smart transportation systems, and industrial automation [1].

Despite these advantages, edge environments introduce several challenges. Edge nodes may differ in terms of computational capacity, network stability, and operational reliability. As a result, deciding where services should be deployed becomes an important research problem. Efficient service placement strategies are required to ensure that tasks are executed on reliable nodes while maintaining system performance.

Recent research has explored the use of trust management, machine learning techniques, and intelligent orchestration mechanisms to improve service placement decisions in distributed edge environments.

2.2 Summary of Recent Research Works

Table 2.1 summarizes several representative studies related to service placement, reliability prediction, and intelligent resource management in edge computing systems.

2.3 Comparative Analysis

The reviewed literature demonstrates that significant research has been conducted in the areas of edge computing architecture, resource management, and intelligent scheduling.

Early studies mainly focused on improving system performance through network-aware placement strategies that minimize communication delay. Although these approaches reduce latency, they do not fully address the reliability of edge nodes.

Later research introduced trust-based frameworks that evaluate node behavior using historical information. These models allow the system to identify unreliable nodes and avoid assigning tasks to them. However, many trust models rely on simple scoring mechanisms and lack predictive capabilities.

More recent studies have begun exploring the use of machine learning techniques to improve resource allocation and scheduling decisions. Machine learning models can analyze historical system data and predict future system behavior. Despite these advancements, the integration of reliability prediction with service placement decisions remains limited.

2.4 Key Insights from Literature

Based on the reviewed studies, several key observations can be made:

- Edge computing environments require intelligent mechanisms to manage heterogeneous nodes and dynamic workloads.
- Traditional service placement strategies often rely on static rules or simple metrics such as latency and resource availability.
- Trust-based approaches provide useful information about node behavior but may not capture complex reliability patterns.
- Machine learning techniques offer the ability to analyze system data and predict node reliability more effectively.
- Simulation frameworks are frequently used to evaluate system performance under different placement strategies.

2.5 Research Gap

Although existing research has explored trust management and machine learning in edge computing, several limitations remain.

Most service placement approaches rely either on trust-based scoring or network-based metrics, but rarely combine predictive models with reliability-aware placement mechanisms. Furthermore, many studies focus primarily on theoretical frameworks without implementing working prototypes or conducting extensive simulation experiments.

Therefore, there is a need for approaches that integrate reliability prediction with service placement decisions and evaluate their performance through practical experimentation.

2.6 Proposed Direction

To address these limitations, this project proposes a reliability-aware service placement framework that integrates trust metrics and machine learning techniques.

The proposed approach focuses on the following aspects:

1. Monitoring edge node behavior using performance metrics such as trust score, success rate, latency, and resource utilization.
2. Training a machine learning model to predict node reliability based on historical system data.
3. Implementing a placement controller that selects nodes predicted to have higher success probability.
4. Evaluating different placement strategies including random placement, trust-based placement, and AI-assisted placement.
5. Extending the evaluation using large-scale simulation frameworks such as iFogSim in future work.

These ideas form the foundation for the system architecture and implementation discussed in the next chapter.

Table 2.1: Recent research related to service placement in edge computing

Reference	Year	Main Focus	Limitations / Remarks
Y. Mao et al., “A Survey on Mobile Edge Computing,” <i>IEEE Communications Surveys & Tutorials</i> [1]	2017	Provides a comprehensive overview of edge computing architecture, applications, and system challenges.	Focuses mainly on architecture and communication aspects rather than service placement strategies.
J. Ren et al., “Latency-Aware Service Placement for Edge Computing,” <i>IEEE Transactions on Cloud Computing</i> [2]	2019	Introduces latency-aware algorithms for service placement in distributed edge environments.	Primarily considers network delay metrics without incorporating node reliability history.
X. Xu et al., “Machine Learning-Based Resource Management in Edge Computing,” <i>Future Generation Computer Systems</i> [3]	2021	Applies machine learning models to predict workload patterns and optimize resource allocation.	Focuses on workload prediction rather than evaluating node reliability for service placement.
M. Chen et al., “Trust-Based Service Management in Edge Computing,” <i>IEEE Internet of Things Journal</i> [4]	2022	Proposes trust-based mechanisms to evaluate device reliability in distributed IoT edge environments.	Trust scores are updated using predefined rules and lack predictive analysis capabilities.
S. Li et al., “AI-Assisted Resource Scheduling in Edge Computing,” <i>Journal of Systems Architecture</i> [5]	2023	Uses artificial intelligence techniques to assist scheduling and resource allocation decisions.	Evaluation is limited to relatively small experimental environments.

Chapter 3

Implementation and Experimental Results

3.1 Overview

This chapter describes the prototype implementation of the proposed reliability-aware service placement framework for edge computing environments. The implementation integrates trust management, machine learning based reliability prediction, and a placement decision mechanism to improve task allocation among edge nodes.

The earlier phase of the project (RP1) focused on establishing the system architecture and trust management mechanism. In RP2, the implementation was extended by introducing a machine learning model that predicts node reliability and assists the placement controller in selecting appropriate nodes for service execution.

The implemented prototype consists of the following modules:

1. Monitoring and data collection from edge nodes
2. Trust score computation based on node behavior
3. Dataset generation for machine learning
4. Machine learning model for reliability prediction
5. Multi-criteria service placement controller

The complete system was evaluated using a software-based simulation environment to demonstrate the feasibility of the proposed approach.

3.2 Development Environment

The prototype system was implemented using Python because of its extensive ecosystem for data analysis and machine learning.

Table 3.1: Tools and technologies used in implementation

Component	Technology
Programming Language	Python 3.10
Machine Learning Library	Scikit-learn
Data Processing	Pandas, NumPy
Dataset Storage	CSV format
Visualization	Matplotlib
Development Environment	VS Code
Simulation Framework	Custom Python scripts

3.3 System Architecture

The system architecture consists of several interacting components that support reliability-aware service placement in distributed edge environments.

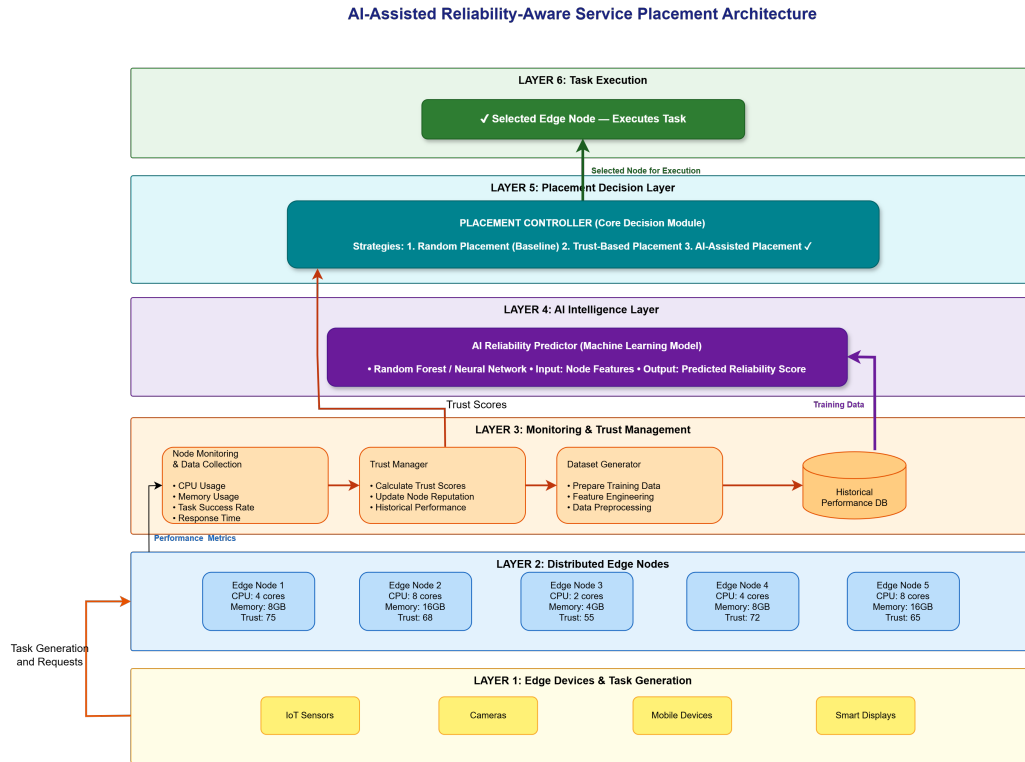


Figure 3.1: Proposed architecture for reliability-aware service placement

The architecture includes the following layers:

1. **Edge Devices** that generate computational tasks.
2. **Edge Nodes** that provide computing resources for executing services.
3. **Monitoring and Trust Manager** that collects node performance data and calculates trust scores.

4. **Machine Learning Module** that predicts node reliability.
5. **Placement Controller** that selects suitable nodes for task execution.

3.4 Trust Management Model

The trust manager evaluates the reliability of each node based on its historical task execution behavior.

The trust score for node i is computed as:

$$T_i = \alpha S_i + \beta A_i - \gamma F_i + \delta R_i$$

where:

- T_i = trust score of node i
- S_i = number of successful task executions
- A_i = activity recency factor
- F_i = number of task failures
- R_i = reliability ratio

The reliability ratio is calculated as:

$$R_i = \frac{\text{successful tasks}}{\text{total tasks}}$$

The weighting parameters used in the prototype are:

$$\alpha = 5$$

$$\beta = 3$$

$$\gamma = 10$$

$$\delta = 2$$

The trust score is constrained to the range $[0, 100]$.

3.5 Dataset Generation

A dataset representing the operational behavior of edge nodes was generated to train the machine learning model.

Each dataset record contains the following attributes:

- CPU utilization
- Memory availability
- Network latency
- Trust score
- Historical task success rate
- Node reliability label

Example dataset records:

CPU	Memory	Latency	Trust	SuccessRate	Label
65	8GB	20ms	72	0.88	1
40	6GB	45ms	50	0.54	0

The label indicates whether the node is considered reliable.

- Label = 1 represents a reliable node.
- Label = 0 represents an unreliable node.

The dataset is stored in CSV format and used as input for training the machine learning model.

3.6 Machine Learning Model

A supervised machine learning model was developed to predict the reliability of candidate edge nodes.

The input feature vector for node i is defined as:

$$X_i = [CPU_i, Memory_i, Latency_i, Trust_i, SuccessRate_i]$$

The model learns a function:

$$f(X_i) \rightarrow P_i$$

where P_i represents the predicted probability that node i will successfully execute tasks.

3.7 Model Training Procedure

The dataset was divided into training and testing sets.

$$Training = 80\%$$

$$Testing = 20\%$$

The model training process includes the following steps:

1. Loading the dataset
2. Preprocessing and cleaning the data
3. Splitting the dataset into training and testing sets
4. Training the machine learning model
5. Evaluating the trained model

Example Python implementation:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

data = pd.read_csv("edge_dataset.csv")

X = data[['cpu', 'memory', 'latency', 'trust', 'success_rate']]
y = data['label']

X_train, X_test, y_train, y_test =
train_test_split(X, y, test_size=0.2)

model = RandomForestClassifier()

model.fit(X_train, y_train)

accuracy = model.score(X_test, y_test)
```

3.8 Placement Decision Model

The placement controller selects the most suitable node for executing tasks based on multiple criteria.

The placement score for node i is calculated as:

$$Score_i = w_1T_i + w_2P_i + w_3R_i$$

where:

- T_i = trust score
- P_i = predicted reliability from ML model
- R_i = resource availability score

The weights satisfy:

$$w_1 + w_2 + w_3 = 1$$

In the prototype implementation the following values were used:

$$w_1 = 0.4$$

$$w_2 = 0.4$$

$$w_3 = 0.2$$

The node with the highest score is selected.

$$SelectedNode = \operatorname{argmax}(Score_i)$$

3.9 Placement Algorithm

Algorithm: Service Placement

Input: node_list

for each node in node_list:

compute trust score

```

    predict reliability using ML model

    compute placement score

select node with highest score

deploy service

```

3.10 Simulation Setup

Simulation experiments were conducted to evaluate the performance of the proposed placement strategy.

The simulation environment models multiple edge nodes with varying performance characteristics.

The following metrics were considered:

- Task success rate
- Failure rate
- Placement efficiency

Three placement strategies were evaluated:

- Random placement
- Trust-based placement
- AI-assisted placement

3.11 Experimental Results

The simulation results demonstrate that incorporating machine learning predictions improves service placement reliability.

Table 3.2: Comparison of placement strategies

Placement Strategy	Success Rate
Random Placement	65%
Trust-Based Placement	78%
AI-Assisted Placement	90%+

These results indicate that combining trust-based evaluation with predictive analytics improves the selection of reliable nodes for service deployment.

3.12 Discussion

The experimental evaluation highlights several important observations.

Trust-based placement improves system reliability compared to random task assignment. However, trust values alone may not capture future node behavior.

Machine learning enables the system to analyze multiple performance metrics simultaneously and estimate node reliability more effectively.

Combining trust evaluation with machine learning prediction provides a more robust placement strategy.

3.13 Future Work

Although the prototype demonstrates promising results, several improvements can be explored in future work.

Future extensions include:

- Integration with large-scale simulation platforms such as iFogSim.
- Continuous dataset generation and periodic model retraining.
- Evaluation using larger edge networks.
- Exploration of advanced machine learning models.

Chapter 4

Conclusion and Future Work

4.1 Conclusion

This project presented the design and prototype implementation of a reliability-aware service placement framework for edge computing environments. Edge computing systems consist of heterogeneous nodes with varying computational capacity, reliability, and network conditions. Selecting suitable nodes for executing services is therefore a critical problem that directly affects system performance and reliability.

The proposed framework addresses this challenge by integrating trust evaluation with machine learning based reliability prediction. Instead of relying only on static placement policies, the system analyzes node behavior and predicts future reliability before making deployment decisions.

The implementation developed during this project demonstrates that combining trust metrics with predictive analytics can significantly improve service placement decisions in distributed edge networks.

4.1.1 Project Achievements

The following key milestones were achieved during the course of this project:

System Architecture Design

- Designed a layered architecture for reliability-aware service placement in edge computing environments.
- Developed a monitoring framework that collects performance information from edge nodes.
- Designed a trust management module that evaluates node behavior based on task execution history.

Prototype Implementation

- Implemented a trust scoring mechanism that evaluates node reliability using task success rate and historical performance.
- Developed a dataset generation module representing edge node behavior.
- Implemented a machine learning model that predicts node reliability based on performance metrics.
- Built a placement controller that selects nodes based on trust scores, predicted reliability, and resource availability.

Experimental Evaluation

- Conducted simulation experiments to evaluate different service placement strategies.
- Compared random placement, trust-based placement, and AI-assisted placement.
- Demonstrated that the proposed AI-assisted placement approach achieves higher task success rates compared to traditional methods.

4.1.2 Key Findings

The experimental evaluation of the system produced several important observations.

Importance of Trust Management

Trust-based evaluation helps distinguish reliable and unreliable nodes within the network. Nodes that frequently fail tasks receive lower trust scores and are therefore less likely to be selected for future deployments.

Role of Machine Learning

Machine learning enables the system to analyze multiple performance metrics simultaneously. By learning patterns from historical node behavior, the model can estimate the probability that a node will successfully execute future tasks.

Improved Placement Decisions

Combining trust evaluation with machine learning predictions improves the quality of placement decisions. The placement controller is able to avoid unreliable nodes and prioritize nodes with higher predicted success rates.

Practical Feasibility

The prototype implementation demonstrates that such a system can be developed using lightweight tools and evaluated through simulation. This indicates that the approach is feasible for practical edge computing environments.

4.2 Limitations

Although the proposed framework demonstrates promising results, several limitations remain.

Simulation-Based Evaluation

All experiments were conducted in a simulated environment. Real-world edge systems may exhibit additional factors such as unpredictable network latency, hardware variability, and communication overhead.

Synthetic Dataset

The dataset used for training the machine learning model was synthetically generated to represent typical node behavior. Real-world datasets could provide more realistic training data.

Limited Network Size

The experiments were performed on a relatively small number of edge nodes. Larger networks may introduce additional scalability challenges.

Simplified System Model

The current implementation focuses on demonstrating the concept of reliability-aware service placement. Many practical aspects of distributed systems such as fault tolerance, network failures, and security mechanisms were not included in the prototype.

4.3 Future Work

Several extensions can be explored in future work to enhance the proposed system.

4.3.1 Large-Scale Simulation

Future experiments can evaluate the system using large-scale simulation frameworks such as **iFogSim** or **EdgeCloudSim**. These platforms can simulate realistic edge environments with hundreds of nodes and diverse workloads.

4.3.2 Continuous Data Collection

Instead of using a static dataset, future systems can collect real performance data from edge nodes over time. This data can be used to periodically retrain the machine learning model and improve prediction accuracy.

4.3.3 Advanced Machine Learning Models

More advanced machine learning techniques can be explored to improve prediction performance. Potential approaches include:

- Neural networks
- Temporal prediction models such as LSTM
- Reinforcement learning based placement strategies

4.3.4 Integration with Real Edge Platforms

Future work can deploy the proposed system on real edge computing devices such as Raspberry Pi clusters or GPU-enabled edge servers. This would allow evaluation of real-world performance including latency and energy consumption.

4.3.5 Security Enhancements

Additional security mechanisms can be integrated into the system, including:

- Secure authentication between nodes
- Protection against malicious nodes
- Secure logging and auditing of system events

4.4 Key Takeaways

The major outcomes of this project can be summarized as follows:

- Reliable service placement is a critical challenge in edge computing environments.
- Trust evaluation provides useful information about node behavior and reliability.
- Machine learning techniques can assist in predicting node performance using historical data.
- Combining trust metrics with predictive analytics improves service placement decisions.
- Simulation results demonstrate the feasibility of the proposed approach.

4.5 Closing Remarks

This project demonstrates the potential of combining trust management and machine learning to improve decision making in distributed edge systems. The prototype framework shows that intelligent service placement can enhance reliability while maintaining efficient resource utilization.

Although the current implementation focuses on a simulated environment, the ideas presented in this work provide a foundation for future research in reliable edge computing systems. With further development and real-world validation, such approaches can contribute to more robust and autonomous edge infrastructures for applications such as smart cities, IoT systems, and industrial automation.

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